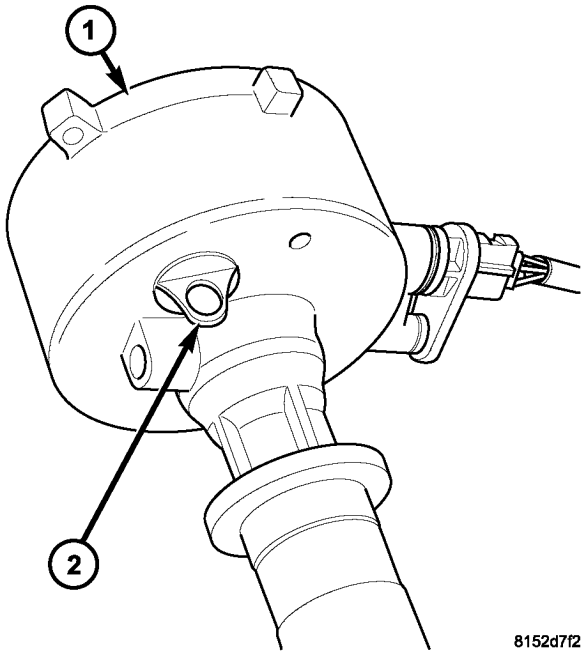

Fig. 20 4.0L CMP FACTORY ALIGNMENT PIN

1 - OIL PUMP DRIVE HOUSING
2 - FACTORY ALIGNMENT PIN

(3) If installing a **new** oil pump drive assembly, it is supplied with a temporary alignment pin (2) (Fig. 20) to prevent the target wheel from rotating. Do not remove this pin until oil pump drive assembly (1) has been installed.

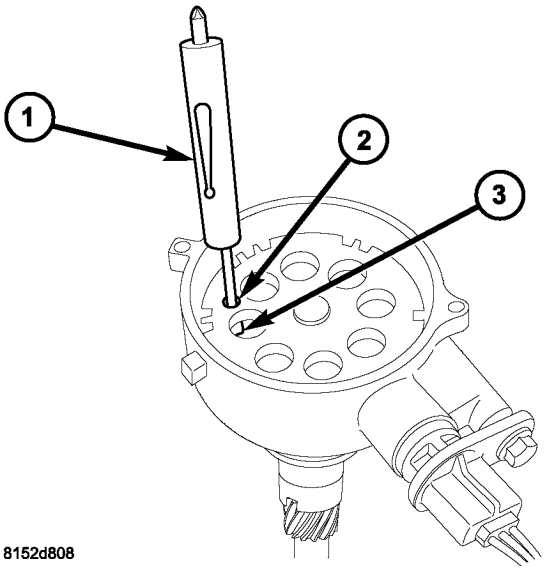

Fig. 21 4.0L CMP TARGET WHEEL ALIGNMENT

1 - SMALL SCREWDRIVER
2 - TARGET WHEEL ALIGNMENT HOLE
3 - HOUSING ALIGNMENT HOLE

(4) If installing/returning a **used** oil pump drive assembly back to the engine, temporarily install a small screwdriver (1) (Fig. 21) or similar tool through target wheel access hole (2) and then through mating hole in housing (3).

(5) Install oil pump drive into engine while aligning into slot on oil pump. Rotate oil pump drive housing back to its original 4 o'clock position as viewed from right side of engine. Install hold-down clamp and bolt. Tighten bolt.

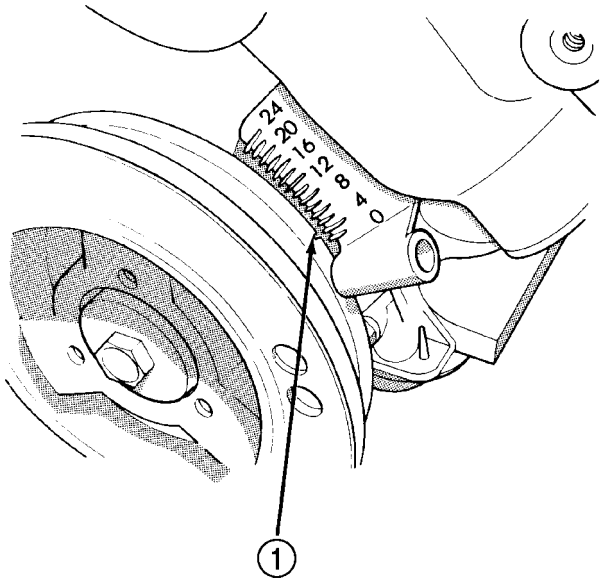

Fig. 22 Align Timing Marks - 4.0L Engine

1 - CRANKSHAFT VIBRATION DAMPER TIMING MARK

(6) If engine crankshaft or camshaft has been rotated, such as during engine tear-down, CMP sensor relationship must be reestablished.

(a) Remove ignition coil rail assembly. Refer to Ignition Coil Removal/Installation.

(b) Remove cylinder number 1 spark plug.

(c) Hold a finger over the open spark plug hole. Rotate engine at vibration dampener bolt until compression (pressure) is felt.

(d) Slowly continue to rotate engine. Do this until timing index mark (1) (Fig. 22) on vibration damper pulley aligns with top dead center (TDC) mark (0 degree) on timing degree scale. Always rotate engine in direction of normal rotation. Do not rotate engine backward to align timing marks.

(e) Install oil pump drive into engine while aligning into slot on oil pump. If pump drive will not drop down flush to engine block, the oil pump slot is not aligned. Remove oil pump drive and align slot in oil pump to shaft at bottom of drive. Install into engine. Rotate oil pump drive housing back to its original 4 o'clock position as viewed from right side of engine. Install hold-down clamp and bolt.